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Sequences (Ch. 11.1)

A sequence is a list of numbers written in a definite order

$$\underset{\text{1st term}}{a_1}, \underset{\text{2nd term}}{a_2}, \underset{\text{3rd term}}{a_3}, \ldots, \underset{\text{nth term / general term}}{a_n}, \ldots$$

We'll usually deal with infinite sequences, where each term a_n has a successor a_{n+1}. And we also denote them as $\{a_n\}$ or $\{a_n\}_{n=1}^{\infty}$ or $\{a_1, a_2, \ldots, a_n, \ldots\}$

n usually starts at 1, but does NOT have to.

Examples:

1) $\left\{\frac{n}{n+1}\right\}_{n=1}^{\infty}$ OR $a_n = \frac{n}{n+1}$ OR $\left\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \ldots\right\}$

2) $\left\{\sqrt{n-3}\right\}_{n=3}^{\infty}$ OR $a_n = \sqrt{n-3}, n \geq 3$ OR $\{0, 1, \sqrt{2}, \sqrt{3}, \ldots\}$

Note that the above sequences are given in an explicit form. However, sometimes they can be given by recursive formulas

Example: $a_1 = 1$, $a_n = 2a_{n-1} + 1$

$\Rightarrow a_2 = 2 \cdot 1 + 1 = 3$, $a_3 = 2 \cdot 3 + 1 = 7$ etc.

Finally, note also that a sequence can be viewed as a function

$$\mathbb{N} \to \mathbb{R}$$
$$n \to a_n$$

Convergence / Divergence

We say that a sequence $\{a_n\}$ converges (or it is convergent) if there exists a number $L \in \mathbb{R}$ s.t.

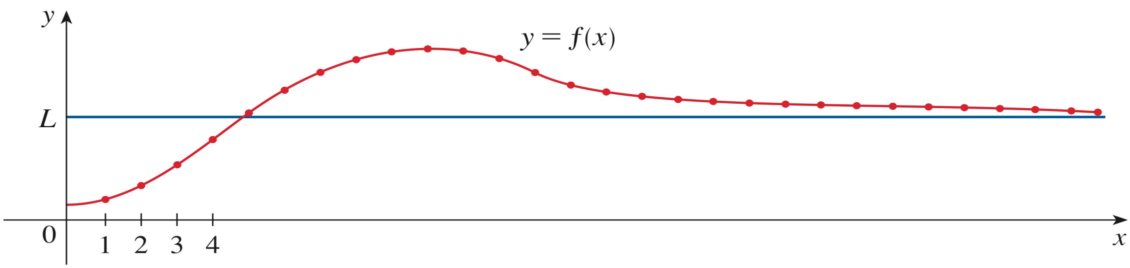
$$\lim_{n\to\infty} a_n = L$$

Otherwise, we say that $\{a_n\}$ diverges (or it is divergent)

Example: Find limit of $a_n = \frac{5n^2-3n+1}{2n^2+7}$ or show it does not exist.

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{\overbrace{\frac{5n^2}{n^2}}^{=5} - \overbrace{\frac{3n}{n^2}}^{\to 0} + \overbrace{\frac{1}{n^2}}^{\to 0}}{\underbrace{\frac{2n^2}{n^2}}_{2} + \underbrace{\frac{7}{n^2}}_{\to 0}} = \boxed{\frac{5}{2}}$$

Note that we have the general rule: if $f(x)$ is a fct. s.t. $a_n = f(n)$, then $\lim_{x\to\infty} f(x) = L \Rightarrow \lim_{n\to\infty} a_n = L$.



For instance, in the previous example $a_n = f(n)$, with $f(x) = \frac{5x^2-3x+1}{2x^2+7}$.

$$\lim_{x\to\infty} \frac{5x^2-3x+1}{2x^2+7} = \frac{\infty}{\infty} \overset{H}{=} \lim_{x\to\infty} \frac{10x-3}{4x} = \frac{\infty}{\infty} \overset{H}{=} \lim_{x\to\infty} \frac{10}{4} = \boxed{\frac{5}{2}}$$

Limit Laws for sequences

Assume $\{a_n\}, \{b_n\}$ are convergent and $c \in \mathbb{R}$. Then

1. $\lim_{n\to\infty} (a_n \pm b_n) = \lim_{n\to\infty} a_n \pm \lim_{n\to\infty} b_n$

2. $\lim_{n\to\infty} (c\,a_n) = c\left(\lim_{n\to\infty} a_n\right)$

3. $\lim_{n\to\infty} (a_n b_n) = \lim_{n\to\infty} a_n \cdot \lim_{n\to\infty} b_n$

4. $\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{\lim_{n\to\infty} a_n}{\lim_{n\to\infty} b_n}$, if $\lim_{n\to\infty} b_n \neq 0$.

5. $\lim_{n\to\infty} a_n^p = \left[\lim_{n\to\infty} a_n\right]^p$, if $p>0$ and $a_n>0$.

Few other properties

1. The Squeeze Theorem: If $a_n \le b_n \le c_n$, for $n \geq n_0$, $\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = L$, then

$$\lim_{n\to\infty} b_n = L.$$

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Example: Determine the convergence or divergence of $\left\{ \frac{\sin n}{n^2} \right\}_{n=1}^{\infty}$.

Sol: $-1 \le \sin n \le 1$ for all $n \ge 1$ $\quad 1 : n^2$

$$\Rightarrow \underbrace{-\frac{1}{n^2}}_{\xrightarrow{n\to\infty} 0} \le \frac{\sin n}{n^2} \le \underbrace{\frac{1}{n^2}}_{\xrightarrow{n\to\infty} 0} \quad \xRightarrow[\text{Th.}]{\text{squeeze}}$$

$$\Rightarrow \lim_{n\to\infty} \frac{\sin n}{n^2} = 0$$

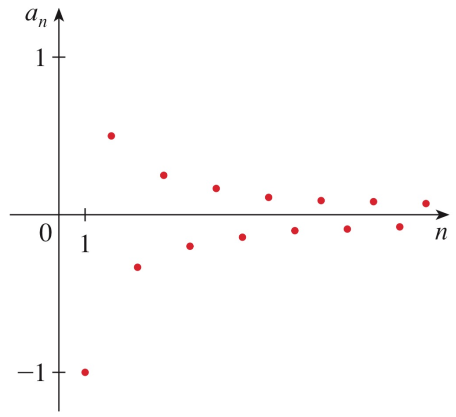
② If $\lim_{n\to\infty} |a_n| = 0$, then $\lim_{n\to\infty} a_n = 0$.

Example: Determine the convergence or divergence of $\left\{ \frac{(-1)^n}{n} \right\}_{n=1}^{\infty}$.

$$\left| \frac{(-1)^n}{n} \right| = \frac{1}{n}$$

$$\lim_{n\to\infty} \left| \frac{(-1)^n}{n} \right| = \lim_{n\to\infty} \frac{1}{n} = 0 \quad \Rightarrow$$

$$\Rightarrow \lim_{n\to\infty} \frac{(-1)^n}{n} = 0.$$



③ If $\lim_{n\to\infty} a_n = L$ and f is cont. at L, then

$$\lim_{n\to\infty} f(a_n) = f(L).$$

Example: Find $\lim_{n\to\infty} \cos\left(\frac{\pi}{n}\right)$

$\lim_{n\to\infty} \frac{\pi}{n} = 0$

$f(x) = \cos x$ is cont at 0

$\Rightarrow \lim_{n\to\infty} \cos\left(\frac{\pi}{n}\right) = \cos 0 = \boxed{1}$

4. $\{r^n\}$ is convergent, if $r \in (-1, 1]$

divergent, if $r \in (-\infty, -1] \cup (1, +\infty)$

Monotonic and Bounded Sequences

We say that a sequence $\{a_n\}$ is

- **increasing** if $a_n < a_{n+1}$, $n \geq 1$
- **decreasing** if $a_n > a_{n+1}$, $n \geq 1$
- **monotonic** if a_n is either decreasing or increasing.
- **bounded above** if $\exists M$ s.t. $a_n \leq M$ for all $n \geq 1$
- **bounded below** if $\exists m$ s.t. $a_n \geq m$ for all $n \geq 1$
- **bounded** $\Leftrightarrow$ bounded above & below

Example: Show that the sequence $a_n = \frac{n}{n^2+1}$ is decreasing.

Sol 1: We have to show that $a_n > a_{n+1} \Leftrightarrow$

$\Leftrightarrow \frac{n}{n^2+1} > \frac{n+1}{(n+1)^2+1} \Leftrightarrow n[\underbrace{(n+1)^2+1}_{n^2+2n+2}] > (n+1).(n^2+1)$

$\Leftrightarrow \cancel{n^3} + 2n^2 + 2n > \cancel{n^3} + n + n^2 + 1$

$\Leftrightarrow n^2 + n > 1$ for $n \geq 1$, which is true

Sol 2: let $f(x) = \frac{x}{x^2+1} \Rightarrow$

$\Rightarrow f'(x) = \frac{1.(x^2+1) - x.2x}{(x^2+1)^2} = \frac{1-x^2}{(x^2+1)^2} < 0$, for $x > 1$

$\Rightarrow f$ is decreasing on $(1, \infty)$

But $a_n = f(n) \Rightarrow a_n$ is decreasing.

Example: Show that the sequence $\{\underbrace{\frac{3-4n^2}{n^2+1}}_{a_n}\}_{n=1}^{\infty}$ is bounded.

Sol: $|a_n| = \left|\frac{3-4n^2}{n^2+1}\right| \overset{\substack{4n^2-3>0 \text{ for } n\geq 1\\ \downarrow}}{=} \frac{4n^2-3}{n^2+1} < \frac{4n^2}{n^2+1}$

$< \frac{4n^2}{n^2} = 4 \Rightarrow -4 < a_n < 4$ for all $n \geq 1$.

so $\{a_n\}$ is bounded.

Remarks:

① Not every bounded sequence is convergent

For instance $a_n = (-1)^n$ is bounded (since $-1 \leq a_n \leq 1$), BUT divergent (its terms are alternating: $-1, 1, -1, 1, \ldots$)

② Not every monotonic sequence is convergent.

For instance $a_n = n$ is increasing, BUT $a_n \to \infty$.

However, we have:

Theorem (monotonic sequence theorem):
Every bounded monotonic sequence converges.

In fact:

- if $\{a_n\}$ is increasing + bounded above, then a_n converges
- if $\{a_n\}$ is decreasing + bounded below, then a_n converges.

Example: Investigate the convergence of

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the sequence $\left\{ \underbrace{\frac{2^n}{n!}}_{a_n} \right\}_{n=1}^{\infty}$

Sol: First let's note it's NOT really clear how to compute $\lim_{n\to\infty} a_n$. And we cannot use L'Hospital rule for the ∞/∞ case.

- monotonicity: $\frac{a_{n+1}}{a_n} = \frac{\frac{\cancel{2^{n+1}}\,2}{\cancel{(n+1)!}\,n+1}}{\frac{\cancel{2^n}}{\cancel{n!}}} = \frac{2}{n+1} \leq 1$ for $n \geq 1$

$\Rightarrow \{a_n\}$ is decreasing

- boundedness: Clearly $a_n \geq 0$, $\forall n \geq 1$

$\Rightarrow \{a_n\}$ is bounded by below

Therefore $\{a_n\}$ must be convergent.

To find the limit we use

$$\frac{a_{n+1}}{a_n} = \frac{2}{n+1} \Leftrightarrow a_{n+1} = \frac{2}{n+1} a_n$$

$$\Rightarrow L = \lim_{n\to\infty} a_{n+1} = \lim_{n\to\infty} \left(\frac{2}{n+1} \cdot a_n \right) =$$

$$= \lim_{n\to\infty} \frac{2}{n+1} \cdot \lim_{n\to\infty} a_n = 0 \cdot L = 0$$

$\Rightarrow \boxed{L = 0}$

Exercises (Ch. 11.1)

3-16 List the first five terms of the sequence

(6) $\left\{ \frac{n^2-1}{n^2+1} \right\}_{n=3}^{\infty}$

$\left\{ \frac{8}{10}, \frac{15}{17}, \frac{24}{26}, \frac{35}{37}, \frac{48}{50}, \dots \right\} = \left\{ \frac{4}{5}, \frac{15}{17}, \frac{12}{13}, \frac{35}{37}, \frac{24}{25}, \dots \right\}$

(11) $a_n = \frac{(-2)^n}{(n+1)!}$

$\left\{ \frac{-2}{2!}, \frac{4}{3!}, \frac{-8}{4!}, \frac{16}{5!}, \frac{-32}{6!}, \dots \right\} = \left\{ -1, \frac{2}{3}, -\frac{1}{3}, \frac{2}{15}, -\frac{2}{45}, \dots \right\}$

17-22 Find a formula for the general term a_n of the sequence, assuming that the pattern of the first few terms continues.

(19) $\left\{ -3, 2, -\frac{4}{3}, \frac{8}{9}, -\frac{16}{27}, \dots \right\}$

$\uparrow a_1 \quad \uparrow a_2 \quad \uparrow a_3 \quad \uparrow a_4 \quad \uparrow a_5$

We note that $a_1 = -3$ and $\frac{a_n}{a_{n-1}} = -\frac{2}{3}$ $\Leftrightarrow$

$\Rightarrow a_2 = -3\left(-\frac{2}{3}\right), \; a_3 = -3\left(-\frac{2}{3}\right)^2$ etc.

$\Rightarrow a_n = -3\left(-\frac{2}{3}\right)^{n-1}$

(21) $\left\{ \frac{1}{2}, -\frac{4}{3}, \frac{9}{4}, -\frac{16}{5}, \frac{25}{6}, \dots \right\}$

$\uparrow a_1 \quad \uparrow a_2 \quad \uparrow a_3 \quad \uparrow a_4 \quad \uparrow a_5$

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$$a_n = (-1)^{n+1} \frac{n^2}{n+1}.$$

$\boxed{27-62}$ Determine whether the sequence converges or diverges. If it converges, find the limit.

29 $a_n = \dfrac{4n^2-3n}{2n^2+1}$.

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{4n^2}{2n^2} = 2 \quad \text{converges}$$

30 $a_n = \dfrac{4n^2-3n}{2n+1}$

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{4n^2}{2n} = \lim_{n\to\infty} 2n = \infty \quad \text{diverges}$$

33 $a_n = 3^n 7^{-n}$

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \left(\underbrace{\frac{3}{7}}_{r=3/7<1.}\right)^n = 0 \quad \text{converges}$$

37 $a_n = \sqrt{\dfrac{1+4n^2}{1+n^2}}$

$$\lim_{n\to\infty} a_n = \sqrt{\lim_{n\to\infty} \frac{1+4n^2}{1+n^2}} = \sqrt{4} = 2 \quad \text{converges}$$

43 $\left\{ \underbrace{\dfrac{(2n-1)!}{(2n+1)!}}_{=a_n} \right\}$

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$$a_n = \frac{\cancel{(2n-1)!}}{\cancel{(2n-1)!}\cdot 2n\cdot(2n+1)} = \frac{1}{2n(2n+1)}$$

$\Rightarrow \lim_{n\to\infty} a_n = 0.$ converges

46 $a_n = \frac{\tan^{-1} n}{n}$

$$-\frac{\pi}{2} < \tan^{-1} n < \frac{\pi}{2} \Rightarrow -\frac{\pi}{2n} < \frac{\tan^{-1} n}{n} < \frac{\pi}{2n}$$

$-\frac{\pi}{2n} \xrightarrow{n\to\infty} 0 \xleftarrow{n\to\infty} \frac{\pi}{2n}$

$\Rightarrow \lim_{n\to\infty} a_n = 0$ converges

52 $a_n = 2^{-n} \cos n\pi$

$$-1 \leq \cos n\pi \leq 1 \Rightarrow -\frac{1}{2^n} < \frac{\cos n\pi}{2^n} < \frac{1}{2^n}$$

$-\frac{1}{2^n} \xrightarrow{n\to\infty} 0 \xleftarrow{n\to\infty} \frac{1}{2^n}$

$\Rightarrow \lim_{n\to\infty} a_n = 0$ converges

$\boxed{78-84}$ Determine whether the sequence is increasing, decreasing or not monotonic. Is the sequence bounded?

78 $a_n = \cos n = f(n)$ where $f(x) = \cos x$

But $f(x) = \cos x$ is NOT monotonic $\Rightarrow$

$\Rightarrow a_n = f(n)$ is NOT monotonic.

However $-1 \le \cos n \le 1 \Rightarrow a_n$ bounded.

80 $a_n = \frac{1-n}{2+n} = f(n)$, where $f(x) = \frac{1-x}{2+x}$

$$\Rightarrow f'(x) = \frac{-1(2+x) - (1-x)\cdot 1}{(2+x)^2} = \frac{-3}{(2+x)^2} < 0$$

$\Rightarrow f$ is decreasing on $\mathbb{R}$

$\Rightarrow a_n = f(n)$ is decreasing.

Clearly $a_n \le 0$, $n \ge 1$

Since $a_n \searrow$, $a_n \ge \lim_{n\to\infty} a_n = -1$ $\Big| \Rightarrow$

$\Rightarrow a_n$ is bounded.